



UNIVERSITY OF PERADENIYA
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE
STA1023/MAT1063- TUTORIAL 3

- 1) The distribution of the amount of gravel (in tons) sold by a particular construction supply company in a given week is a continuous random variable X with pdf;

$$f(x) = \begin{cases} k(1 - x^2) & ; 0 \leq x \leq 1 \\ 0 & ; \text{otherwise} \end{cases}$$

- Show that the value of k must be $\frac{3}{2}$ in order to have a legitimate pdf.
 - Calculate $E(X)$ and $Var(X)$.
 - Calculate $E(3X - 4)$ and $Var(3X - 4)$.
 - Find the cdf of the random variable X .
 - Use the cdf to calculate the median of X .
- 2) A survey was conducted for 250 students to record the number of hours they spend in the library per week. 75 said 2 hours, 90 said 5 hours and the rest said 6 hours. Let X be the random variable denoted for the number of hours spent in the library per week.
- Derive the pmf of X and the CDF of X .
 - If a student is randomly selected, what is the probability that he/she spend more than 3 hours?
 - If a student is randomly selected, what is the probability that he/she spend less than 5 hours?
 - Find $P(3 \leq X < 6)$.
 - Calculate $E(X)$ & $E(2X + 1)$, $Var(X)$ and $Var(2X + 1)$

- 3) Suppose the cumulative distribution function of X is given by;

$$f(x) = \begin{cases} 0 & ; x < 0 \\ \frac{x}{4} & ; 0 \leq x < 1 \\ \frac{1}{2} + \frac{x-1}{4} & ; 1 \leq x < 2 \\ \frac{11}{12} & ; 2 \leq x < 3 \\ 1 & ; x \geq 3 \end{cases}$$

- a) Find $P(X = i), i = 1, 2, 3$

- b) Find $P(X \geq 2)$.

4) Suppose the MGF of a random variable X is $M_X(t) = (1 - 2t)^{-10}$. Calculate the mean and the variance of X .

5) A random variable X follows the pdf,

$$f(x) = \begin{cases} 1 & ; 0 \leq x \leq 1 \\ 0 & ; \text{o/w} \end{cases}$$

a) Derive the MGF of X .

b) Calculate $E(X)$ and $Var(X)$.

6) A continuous random variable X follows the following pdf.

$$f(x) = \begin{cases} \frac{x}{4} & ; 1 \leq x \leq 3 \\ 0 & ; \text{o/w} \end{cases}$$

a) Find the lower bound for $P(|x - \mu|) < 2\sigma$ using Chebyshev's inequality.

b) Find the exact probability by calculating μ and σ

Submit on or before 2026/08/24 on or before 4.00 pm.